

Department Seminar

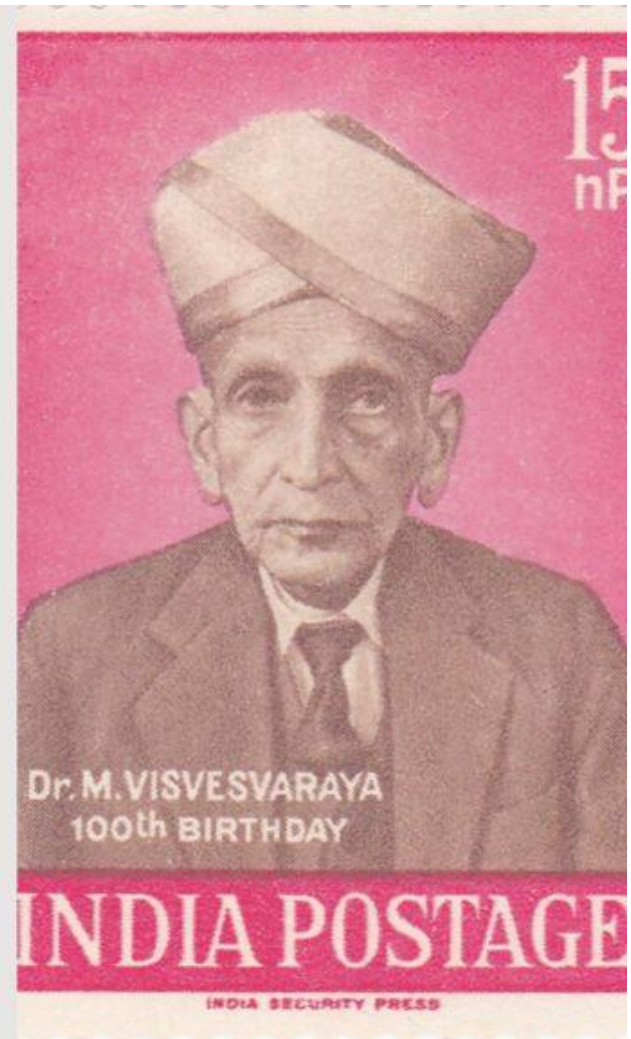


Mahesh Mohan M R
Department of AI
Indian Institute of Technology Kharagpur

"It is not enough to be an
engineer; one must also be an
innovator."

- Sir Mokshagundam Visvesvaraya

Happy Engineers' Day

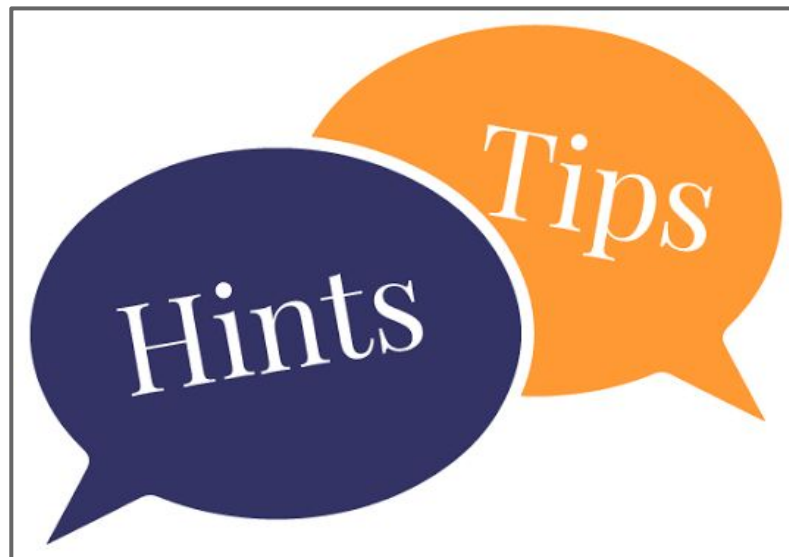
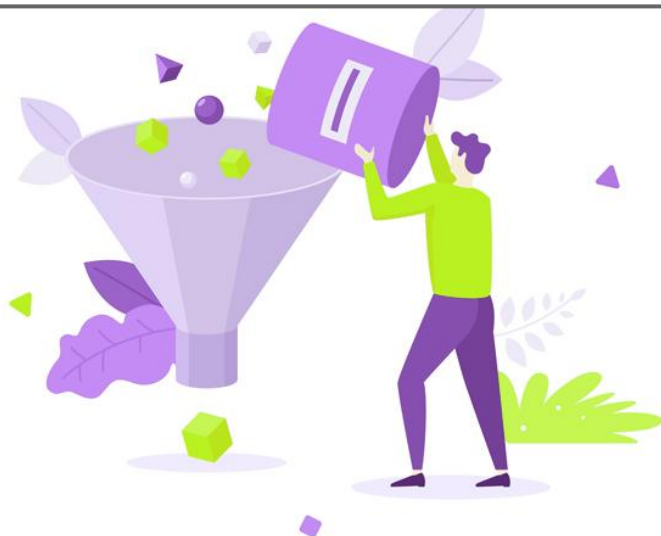




Hints

Yaser S. Abu-Mostafa 1995

California Institute of Technology, Pasadena, CA 91125 USA



ESG Geometry for Sustainability, ICML 26



Uphold transparent management
& ethical business conduct



Reduce carbon emissions,
E-wastes, etc

ESG



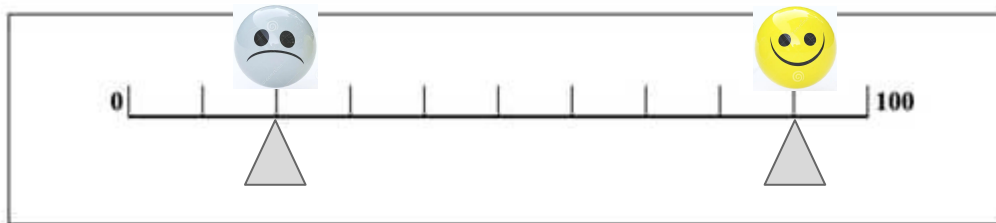
Promote diversity and
labor practices

Firms having higher ESG scores tend to be sustainable in a long term.

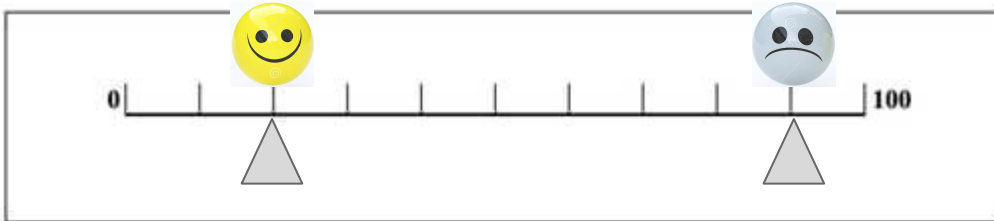
Motivation: Domain Awareness of Tokens

Where Do Multimodal Tokenizers Struggle?

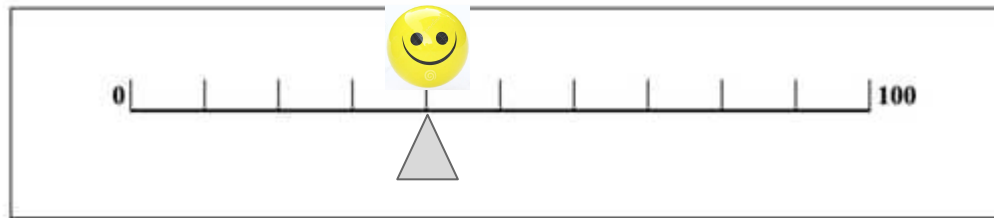
Returns



ESG Risk



Gov Risk



Motivation: Numerical Awareness of Tokens

Where Do Numerical Tokenizers Struggle?

Digit Chunking

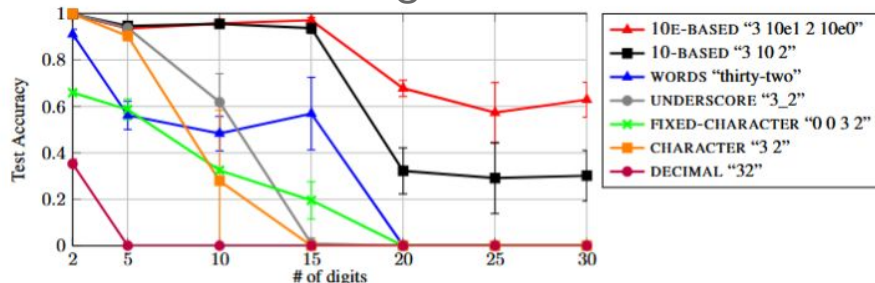
Input → Token IDs
480 481 482 → 22148 11, 4764, 16, 11, 4764, 17

Floating Point Number

"3.14159": "3" + "." + "14" + "159".

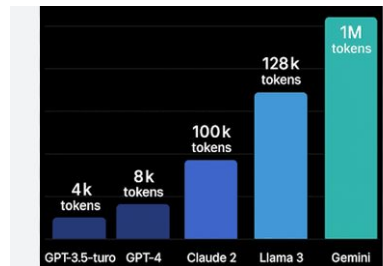
Input → Token IDs
3.14159 → 18, 13, 1415, 19707

Arithmetic Reasoning



Model Specific Behaviour

Not universally consistent across different LLMs



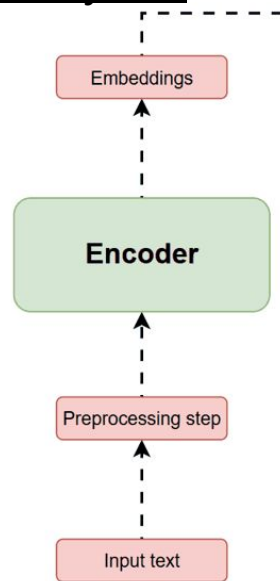
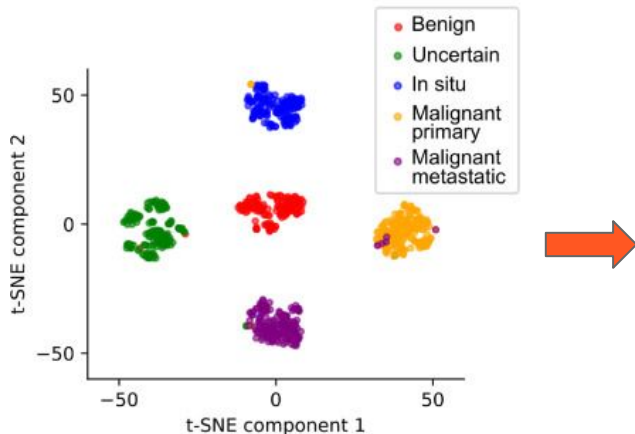
Motivation: Meta-Learning in Foundation Models

- The power of FMs lies in their capacity to learn highly expressive representations that can be adapted to a broad spectrum of tasks.
- FMs typically require additional training stages to become effective for downstream applications → **Data Scarcity; Resource Constraints.**
- Meta-learning helps a model to easily adapt to unseen tasks – largely attributed to their ability to learn useful representations.



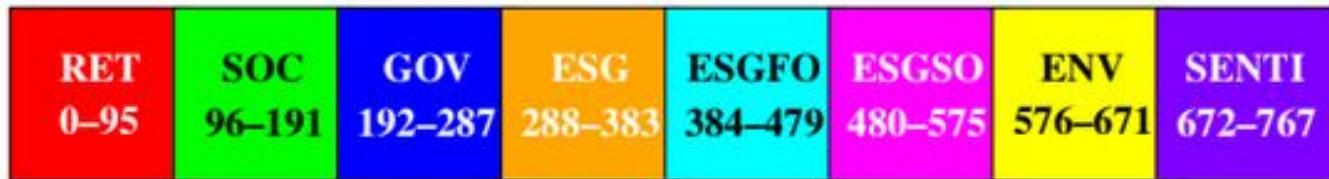
Foundation of the Work

- LLMs form organized structures within their input embedding space and input tokens tend to cluster by type.
- When LLMs are fine-tuned with relationally similar input embeddings it outperforms most of the state of the art methods.
- LLM is capable of modelling relations that go well beyond what the model has seen during training.



Domain Aware Token Embeddings

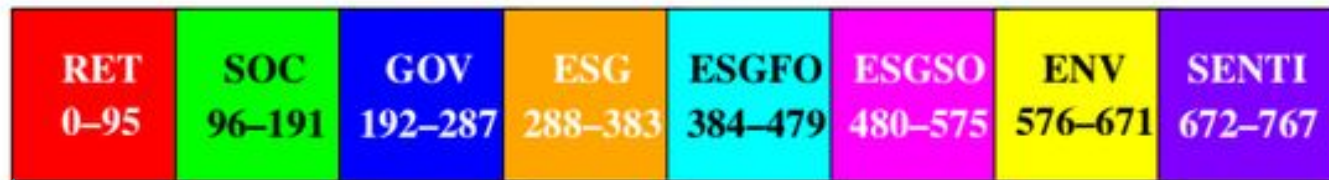
- We encode all numerical variables as typed tokens of the form <TYPE VALUE>. E.g., <ESG 56.04> for ESG risk, <RET 12> for financial return.
- Given a token of the form <TYPE value>, a function maps it to a fixed-dimensional vector $v \in \mathbb{R}$ using orthogonal, block-wise embedding.
- The token prefix (e.g., ESG, RET, etc.) determines the block assignment.



Embedding Space partitioned as per domain

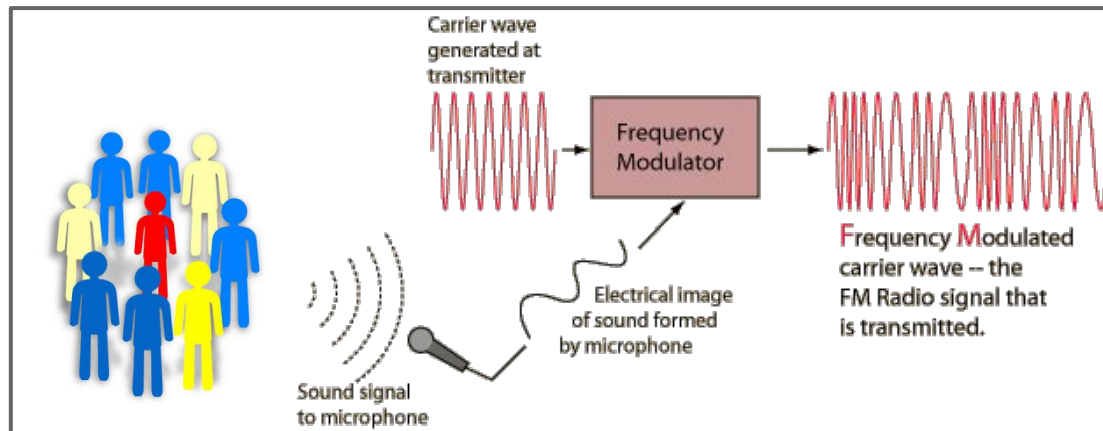
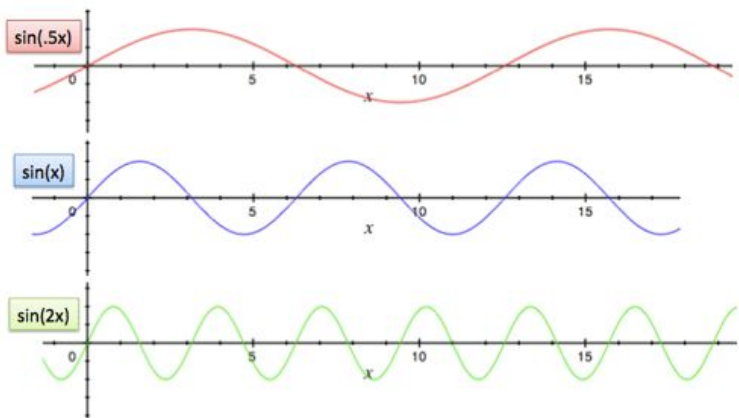
Value Aware Token Embeddings

- If two values x and y are of the same type Q and are numerically close, then the embeddings of $\langle Q \ x \rangle$ and $\langle Q \ y \rangle$ should also be similar.



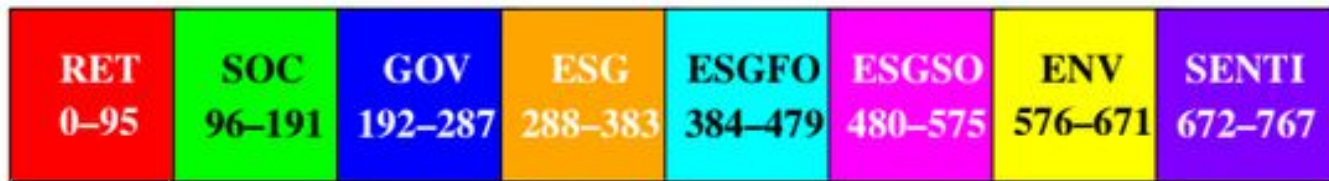
Frequency range (coarse-control)

Growing exponential function for frequency steps (fine-control)



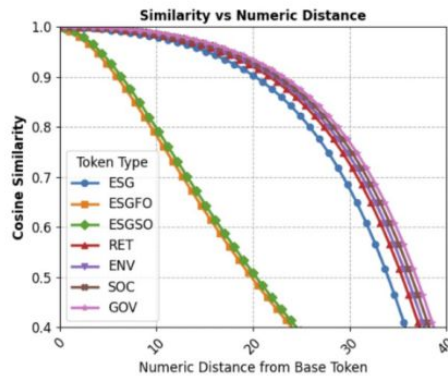
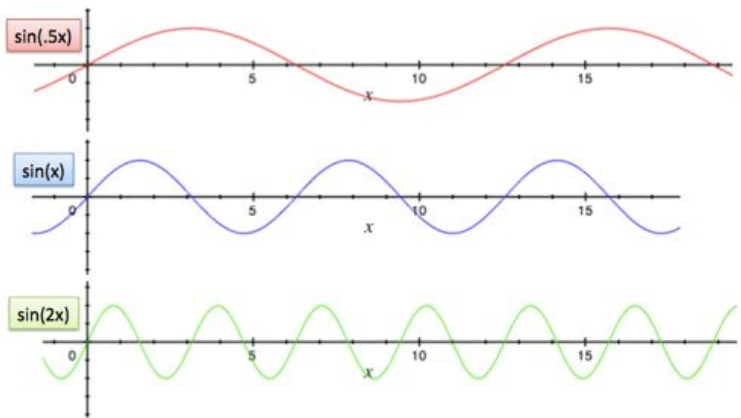
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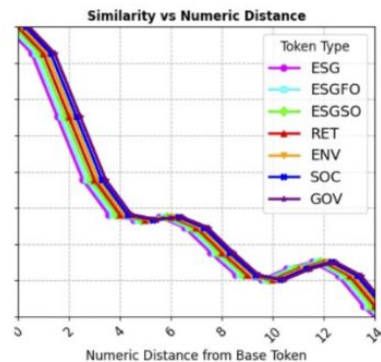


Frequency range (coarse-range)

Quadratic function for frequency steps (fine-range)



Ours



Positional encoding

Contextual Semantics of the Trajectories

$$\mathcal{L} = \mathcal{L}_{\text{forecast}} + \beta \cdot \mathcal{L}_{\text{geometry}},$$

where the first term is the standard LLM token prediction loss, and the second term encourages the model to match distances in the embedding space.

$$\mathcal{L}_{\text{geometry}} = \frac{1}{B} \sum_{b=1}^B w_b \cdot \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \left(D_{ij}^{(b)} - D_{ij}^{\text{DTW}} \right)^2$$

Iterating over embedding blocks

Iterating over
mini-batch
(pair-wise)



Contextual Semantics of the Trajectories

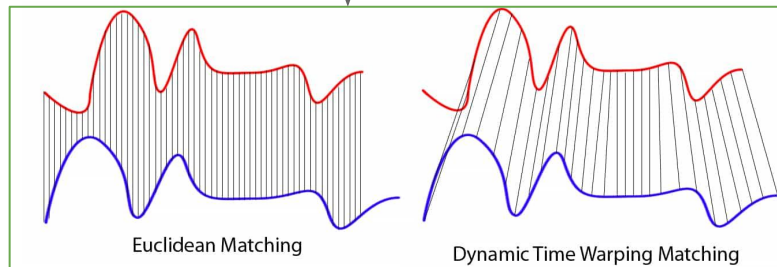
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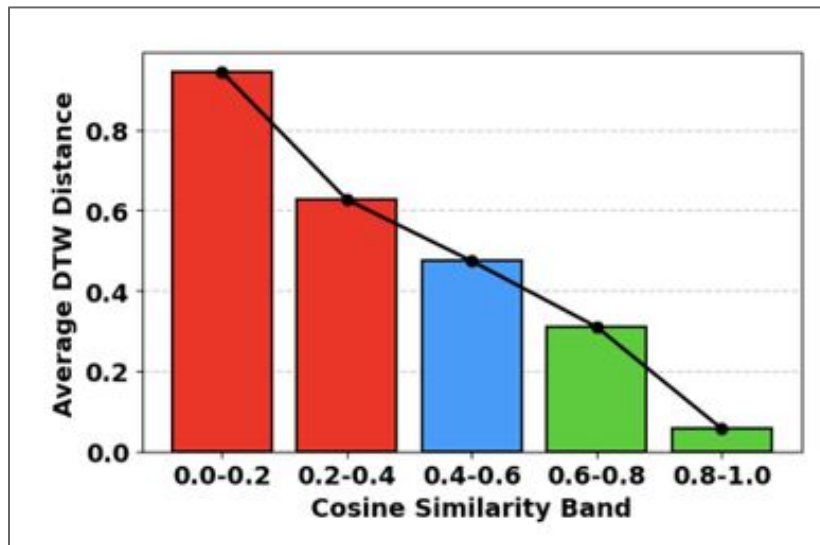
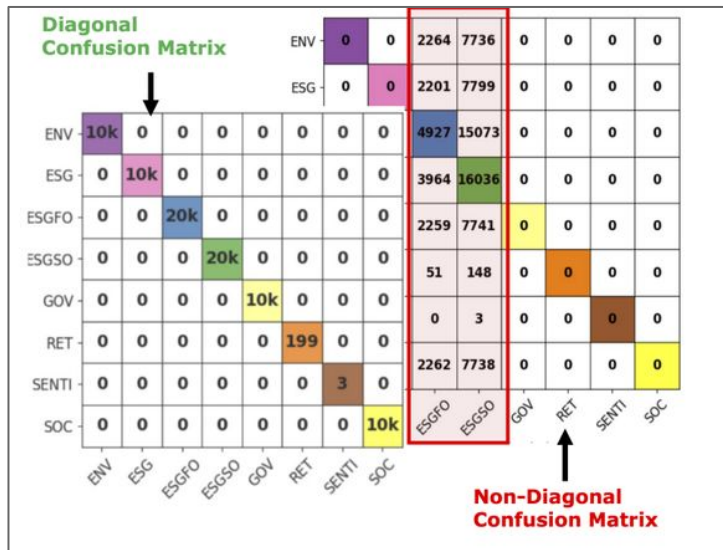


The block-wise geometry loss encourages the learned embedding distances to preserve the structure of the DTW distances.

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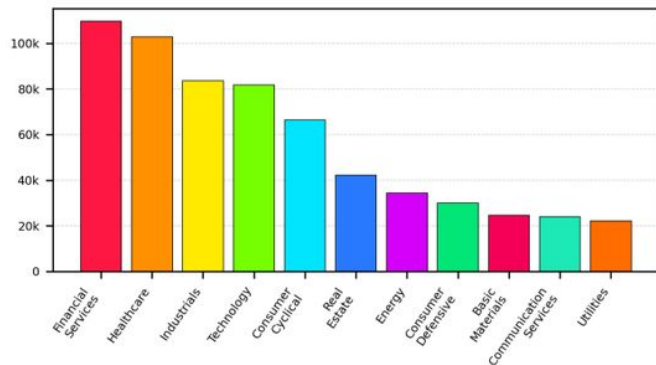
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Relevance of VADT and Geometry Loss

Setting	MSE (↓)	MAE (↓)
Full model (all components)	5.227	1.912
No geometry loss	9.321	2.765
No VADT	128.97	10.53



Batch	Without DTW (s)	With DTW (s)	ΔT (s)	Overhead (%)
32	1.606	1.779	0.173	10.77
64	2.145	2.322	0.177	8.25
128	2.684	2.865	0.181	6.74
256	4.301	4.487	0.186	4.32

Meta-Learning: Superior Asset Population

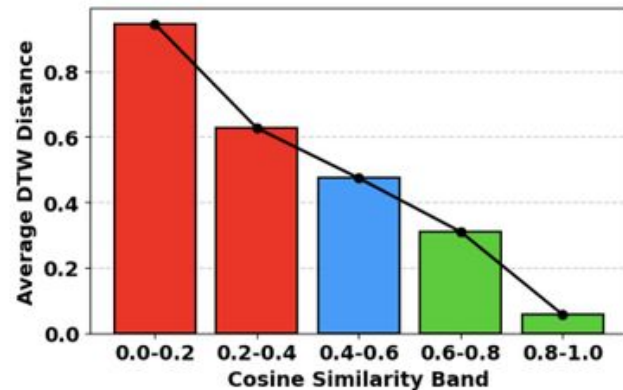
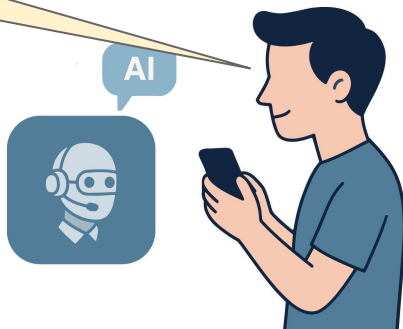
The purpose of SAP is to identify financially similar companies while enabling ESG improvement without sacrificing returns.

$$SAP(x) = \left\{ c \in \mathcal{C} \mid ESG_c < ESG_x \right\}.$$

Cosine similarity with companies of better ESG

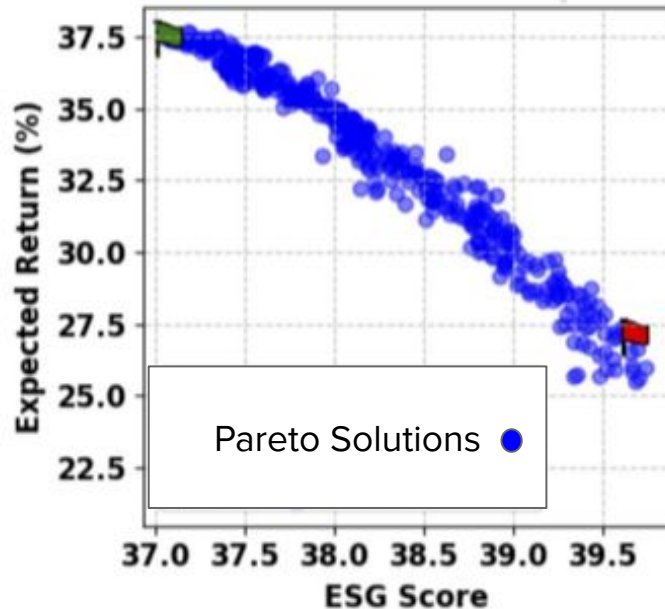
All companies with same trajectory as x

Given a company X , retrieve SAP based on ESG.

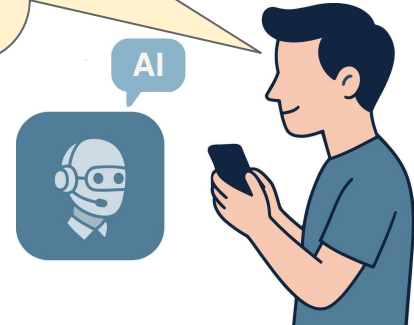


Meta-Learning: Conflicting Objectives

- Multi-objective optimization aims to find a set of solutions that represent the best trade-offs among the objectives
- Pareto front, where no single solution can be considered better than another without improving at least one objective at the expense of another.



Given a company X, retrieve company that exhibit financial return trajectories similar to X, while exhibiting ESG trajectories similar to company Z.

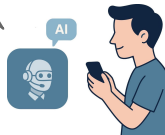


Limitations

- Scaling up in Attribute Level (lacks broader perspectives)

RET	SOC	GOV	ESG	ESGFO	ESGSO	ENV	SENTI
0-95	96-191	192-287	288-383	384-479	480-575	576-671	672-767

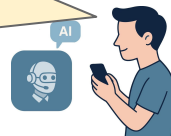
A company lower than AAPL in BUSINESS_PROFILE AND higher than ABBV in CASH_RATIO.



- Proximity vs orderedness

$$\mathcal{L}_{\text{geometry}} = \frac{1}{B} \sum_{b=1}^B w_b \cdot \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N (D_{ij}^{(b)} - D_{ij}^{\text{DTW}})^2$$

Ok, company AAPL is closer to ABBV in ENV. Is it greater or lesser?



- Counterfactual Reasoning

What if Company AAPL changes its carbon emission to 0.65, what will be the new rank?

